I want to design the future of Responsible AI Systems that facilitate collaborative, equitable, and reliable Human-AI Interactions. As a woman from an Indian middle-class background, I have experienced firsthand how socio-psychological factors, such as *community* and *identity*, influence human interactions with technology. This dynamic has fueled my motivation to explore how AI, in its increasing pervasiveness, can be designed to support users from diverse backgrounds and needs. To this end, my work at institutions like **Adobe, IIIT Delhi,** and **Cornell Tech** has centered around tackling *Human-AI Interaction* challenges, particularly in the social contexts of education, equity, and safety. Through **mixed-method empirical research** and **interaction design** techniques, I've developed frameworks and tools to enhance user agency and alignment in AI systems. I aspire to expand on these themes and methods through a **PhD in Computer Science and Engineering at the University of Michigan**. Specifically, I want to answer questions like - **(1) How can we build equitable and safe Human-AI interactions that align with diverse community values? (2) How can we evaluate and improve collaborative Human-AI experiences to foster user trust and agency?**

My doctoral goals are shaped by a three-year journey in HCI that began at IIIT Delhi. The undergraduate HCI coursework, which integrated design, technology, society, and psychology, deeply resonated with my curiosity of the influence of **social dynamics** and **identities** on human interactions. This experience helped me see HCI as a bridge between society and technology, inspiring me to pursue diverse research across **social computing, design,** and **HCI**.

Investigating HCI for Inclusion and Culture: [RE1] My research journey began when I collaborated with **Dr. Aman Parnami** at the **Weave Lab** to develop *Magical Mirrors*, an **AR educational game** for Indian preschoolers. Recognizing that children use mirror play for self-awareness, I designed, developed, and evaluated a game that digitally replicates this learning, incorporating insights from child psychology, qualitative research with parent, and elements from Indian folktales for a culturally relevant storytelling experience. This work, which produced interactive mobile and web games, led to two first-author [1] publications at **IndiaHCI'22** and **Digra'22**. Working on *Magical Mirrors* was pivotal in shaping my current research values, emphasizing empathy for the end user and the integration of qualitative insights into the design of digital experiences that align with user values.

Furthermore, this experience deepened my appreciation for HCI research methods that go beyond quantifiable human experiences and made me perceptive of human-computer interactions in everyday life. [RE2] I observed that many technical systems are designed with users from the **Global North** in focus, at times disregarding the diverse ways these platforms are used globally. At the **Melange Lab** with **Dr. Pushpendra Singh**, I further explored this by studying how the youth in Kashmir—affected by prolonged socio-political conflict and curfews—navigate **digital job-seeking** and **mental health services**. Working alongside Asra Wani (PhD student) for nearly 2 years, I analyzed qualitative interview data and identified critical barriers in their interaction with technology, such as distrust in online platforms, low digital literacy, and unsafe online and algorithmic experiences. I then proposed **inclusive design recommendations** for equitable, safe services based on resilient infrastructures and community values. This experience deepened my understanding of **community-centric technology design** and fueled my goal to study **fairness** and **equity** in socio-technical systems. Our findings were published as co-authored papers at **CSCW'2024**[2] and **CHI'2024**[3], offering valuable experience in academic writing, qualitative research methods, and strong long-term collaborations.

Subsequently, the intersection of **society**, **design**, and **technology** became a central theme in my work. I complemented this focus with specialized coursework in interaction design and computing, HCI research ethics and methods, social psychology, and advanced programming.

Advancing Human-AI Collaboration for User Agency and Equity: My experiences investigating HCI from a cultural and inclusivity lens honed my ability to identify and address challenges that arise when emerging technology intersects with diverse, often overlooked communities.

[RE3,4] One event that captured my attention was the rapid adoption of large language models (LLMs) by my undergraduate peers and professors, particularly in computer science coursework. Curious about the impact of

[1] https://dl.digra.org/index.php/dl/article/view/1322
[2] https://doi.org/10.1145/3637375
[3] https://doi.org/10.1145/3613904.3642507

LLMs on **education in underrepresented communities**, I led a study on LLM usage in **Indian CS classrooms** with **Dr. Dhruv Kumar**. I designed the research methodology, contributed to a dataset of questions tailored to the Indian curriculum, and led a mixed-methods study with students and professors to explore the potential and challenges of using Generative AI in education. The study revealed that while LLMs were versatile, they performed inconsistently and poorly on India-specific queries, highlighting their limitations as educational tools in the Indian context. Building on this, I collaborated with Dr. **Tapan Parikh** at **Cornell Tech** to investigate **implicit gender biases** in LLMs, focusing on underrepresented languages like Hindi. I led the development of a novel method for analyzing Hindi text generation that uncovered patterns of social biases in LLMs, particularly around gender roles, social hierarchies, and class. Both these projects revealed to me the inequitable nature of LLMs, despite their global popularity. Delving into these community-focused issues has contributed greatly to my experience in examining emerging **Human-AI collaborations** through the lens of **user agency** and **inclusion**. Leading these initiatives strengthened my independent research skills, resulting in first-author publications at **SIGCSE'2024**[4] and **ACM ACE'2024**[5] and earning me the **Summer Undergraduate Research Fellowship 2023** at IIITD.

[RE5] Currently, as an **HCI Research Associate** at **Adobe**, I focus on Human-AI collaboration, working in an ML-driven team. As the **first and only HCI researcher** in my team, I've worked independently to integrate human-centered design into our workflows, building bridges between HCI and ML research. My efforts are enhancing my adaptability and skills in cross-domain communications. I am also leading research on Human-AI alignment challenges in the context of **prompt engineering** for **application development**. I am developing **CoPrompter**[6], a tool designed to streamline the identification of misalignments between prompt instructions and model responses, simplifying the manual process of improving prompts to achieve better alignment.

University of Michigan CSE and Future Goals:

My research and lived experiences have exposed critical gaps in Human-AI collaboration that amplify inequities and limit user agency and safety. Challenges in Human AI interactions like, (i) Opacity and propagation of harmful biases in Human AI communication (ii) limited global AI literacy, and (iii) misalignment of model responses with diverse user values, become increasingly critical as AI adoption grows among users with diverse social-cultural backgrounds. Through my PhD, I aim to address these challenges by examining how different communities interact with AI, ultimately designing tools and frameworks to enable inclusive prosocial collaborations. Furthermore, I am eager to expand my research perspectives and build on my core skills in designing and developing interactive systems, informed by insights from mixed-method research.

The Computer Science and Engineering graduate program at the University of Michigan is an ideal fit for me, with its emphasis on interdisciplinary approaches in shaping the future of responsible technologies. Moreover, its emphasis on fostering a diverse and inclusive ecosystem to further cutting-edge research in HCI, Social Computing, and Responsible AI aligns closely with my interests. I am also drawn by the impactful research by the ambitious students and faculty at the CSE department. I am particularly interested **Dr. Q. Vera Liao's** elaborate and significant scholarship on Human-Centered Explainable AI with a focus on building toolkits and evaluation methods to enable user agency and awareness in Human AI Interactions. My research and aspirations closely align with Dr. Vera's visions and I would like to explore interactions to support explainability and AI literacy for non-AI expert users across demographics. Similarly, I am interested in building upon **Dr. Xu Wang's** work on assisting educators and students in building learning strategies and content by leveraging and evaluating responsible AI-enabled tools and frameworks. I am also drawn to **Dr. Farnaz Jahanbakhshi's** work on the social propagation of misinformation through digital platforms and I am keen to explore the role of Generative AI as a medium of mitigating misinformation *or* propagating it. Ultimately, I aspire to advance HCI research as a professor or industrial researcher. A Ph.D. at the University of Michigan will provide me with the skills and knowledge to turn this vision into a reality while also nurturing my growth as a critical thinker and researcher committed to making a lasting impact in the field.

[4] https://doi.org/10.1145/3626252.3630803
[5] https://doi.org/10.1145/3636243.3636257
[6] https://arxiv.org/abs/2411.06099v1